

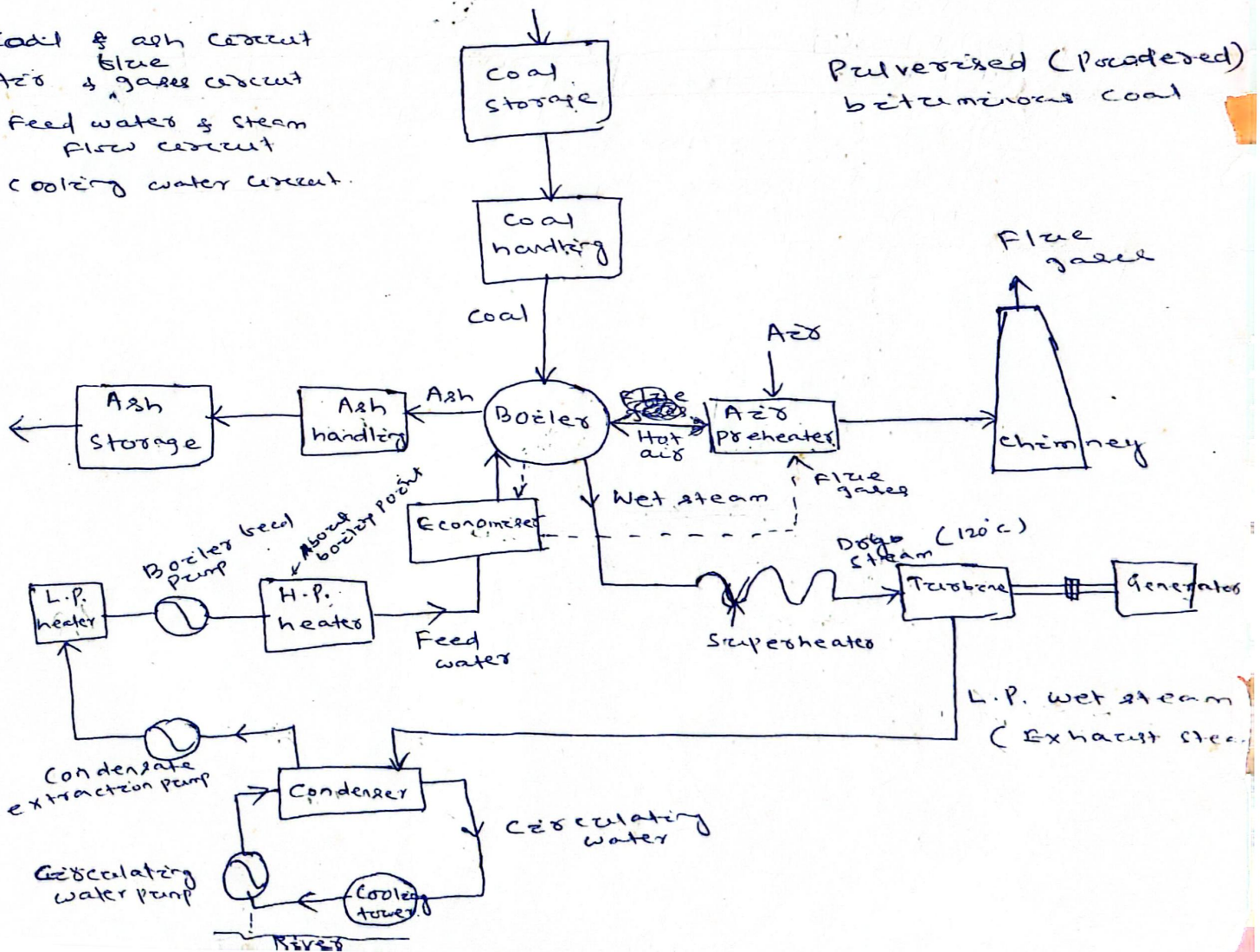
Thermal Power plant (Steam Power plant)

Selection of site for steam power plant

- (1) Availability of raw material (Coal) → 500 MW plant requires about 500-600 tonnes of coal per day.
- (2) Availability of land → 1500 MW requires 90 to 200 acre site
 should be level
 ↓
 should not be rocky or marshy
 ↓
 require blasting ↓
 Pile foundation
- (3) Cost of land
- (4) Availability of water
- (5) Transport facilities
- (6) ^{Dust} Ash disposal facilities → 100 MW unit produces about 400 to 450 tonnes of fly ash per day or 100 tonnes of bottom ash
- (7) Availability of labour (Unskilled and skilled)
- (8) Size of the plant
- (9) Load centre and transmission
- (10) Public problems → Chimney height → 50-60 ft
 Boiler house structure height → 200 ft
- (11) Future extensions

Plant layout (Schematic diagram)

- (1) Coal & ash circuit
- (2) Air & gases circuit
- (3) Feed water & steam flow circuit
- (4) Cooling water circuit



Quesada Components of Thermal power plant

Boiler (Steam generator)

It is a device for generating steam for power, processing or heating purposes.

Boilers are classified on the basis of any of the following characteristics

- (1) Use ☐ Stationary
☐ Mobile
- (2) Pressure
- (3) Materials
- (4) Size
- (5) Tube content ☐ Fire tube
☐ Water tube
- (6) Tube shape & position
- (7) Feeding
- (8) Heat source ☐ Waste heat
☐ Fuel
- (9) Fuel
- (10) Fluid
- (11) Circulation ☐ Natural
☐ Forced
- (12) Furnace position
- (13) Furnace type
- (14) etc.